

RISHABH TIWARI

Boston, MA

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EDUCATION

NorthEastern University
Masters of Science in Computer Science

Boston, MA
Present

Drs. Kiran and Pallavi Global University
Bachelor of Technology in Computer Science, CGPA - 8.83

Vadodara, GJ
Sept 2021 – May 2025

EXPERIENCE

Web Development Intern

Jan 2025 – March 2025

Cognifyz Technology

Remote

- Designed **MongoDB schema** with compound indexes for user profiles, reducing query latency by **40%** on frequently accessed endpoints.
- Built **RESTful API with 15+ Express.js endpoints** for user management, implementing **Joi validation**, **JWT authentication**, and cursor-based pagination.
- Implemented authentication system with **refresh token rotation** and **Redis session storage** to manage user sessions.
- Maintained **90% code coverage** through **Jest** unit and integration tests with automated **CI/CD pipeline**.

Machine Learning Intern

August 2024 – September 2024

CodSoft

Remote

- Built **Flask REST API** serving ML predictions with **95% accuracy**, optimizing response time from **2s to 200ms** through model improvements.
- Developed data preprocessing pipelines using **Pandas**, improving model accuracy by **12%** through feature engineering.
- Implemented **Redis caching layer** to reduce prediction latency by **70%** with **80% cache hit ratio**.
- Set up logging and monitoring to track API latency, throughput, and error rates.

PROJECTS

Multi-Calendar System

Spring Boot, React, MySQL, Redis

- Migrated a legacy Java Swing application to **Spring Boot microservices on AWS**, serving 20+ beta users with **99.9% uptime** during the rollout.
- Implemented **Redis caching** for analytics queries, reducing temporal aggregation latency from **250ms to <1ms**.
- Set up full-stack observability with **Prometheus and Grafana**, tracking **p95 latency** and establishing **SLOs** to identify production performance issues.
- Handled two-way synchronization with the **Google Calendar API** using an **eventual consistency model** and sync tokens to keep data accurate across **600+ IANA timezones**.

ParkEasy Platform

Node.js, Express, MySQL, Redis, WebSockets

- Developed a relational reservation engine with **15+ tables**, managing complex dependencies between users, tiered membership discounts, and dynamic spot availability.
- Engineered **multi-table booking transactions** to ensure data consistency across wallet balances, spot status, and payment records during the **8-step reservation workflow**.
- Implemented concurrency control using **Optimistic Locking** and **WebSockets** to manage real-time parking spot availability and prevent double-bookings.
- Integrated production-grade security features, including **Two-Factor Authentication (2FA)** via Speakeasy, **Role-Based Access Control (RBAC)**, and **JWT-based session management**.
- Implemented testing with **75% code coverage** using **Jest and Supertest**, ensuring reliable error handling.

TECHNICAL SKILLS

Languages: Java, JavaScript, Python, TypeScript, SQL

Frameworks & Libraries: Spring Boot, React, Node.js, Express, Flask, Pandas

Databases & Tools: MySQL, MongoDB, Redis, PostgreSQL

Technologies: REST APIs, WebSockets, JWT, Docker, Git, AWS, Prometheus, Grafana